

Daniele Fossemò

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Bio

Daniele Fossemò is a Ph.D. student at the University of L'Aquila and a member of the AIIMLab research group since 2024, the year he obtained his M.Sc. in Computer Science from the same university cum laude. His research addresses Explainable AI as a foundation for trustworthy machine learning, focusing on Mechanistic Interpretability and on symbolic approaches based on Inductive Logic Programming (ILASP). He has published in *Theory and Practice of Logic Programming* and at BEWARE 2022, and co-organized WEB&GRAPH 2026 workshop, co-located with WSDM 2026.



Education

- 2024 – **Ph.D.**, Computer Science, University of L'Aquila, *on going*
- 2021 – 24 **M.Sc.**, Computer Science (LM-18), University of L'Aquila, 110/110 cum Laude, 20/07/2024.
Thesis: “*Inductive Logic Programming for approximating neural network in Preference Learning*”.
- 2017 – 21 **B.Sc.** Computer Science (L-31), University of L'Aquila, 107/110, 24/07/2021.
Thesis: “*Comparison of machine learning approaches for a food preference system*”.

Schools & Additional Training

- 2026 ECML PKDD 2026, *European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases*. Naples, Italy.
- 2025 *8th Advanced Course on Data Science & Machine Learning (ACDL 2025)*: from Deep Learning to Foundation Models. Riva del Sole Resort, Castiglione della Pescaia (Grosseto), Italy. 8 ECTS.

Research Experience

- 2024 – **Ph.D Student** (XL cycle), University of L'Aquila.

Academic Service

Organization

2025 **Co-organizer** of *Workshop on Web & Graphs, Responsible Intelligence, and Social Media* (WEB&GRAPH 2026), WSDM (*The 19th ACM International Conference on Web Search and Data Mining*) 2026, Boise, Idaho, USA.

Reviewer

2026 **PC member** for *The 35th ACM International Conference of Information and Knowledge Management* (CIKM 26)

2025 **PC member** for *Workshop on Web & Graphs, Responsible Intelligence, and Social Media* (WEB&GRAPH 2026), WSDM (*The 19th ACM International Conference on Web Search and Data Mining*) 2026, Boise, Idaho, USA.

Teaching

University of L'Aquila

2025, 2026 Assistance During Lectures, Deep Neural Networks (Computer Science Msc, LM-18), 8 hours

2024, 2025 Assistance During Lectures, Deep Neural Networks (Computer Science Msc, LM-18), 8 hours

Publications

Journal Articles

- J1. **Fossemò, Daniele**, Mignosi, F., Placidi, G., Raggioli, L., Spezialetti, M. & D'ASARO, F. A. Explaining Neural Networks in Preference Learning: A Post Hoc Inductive Logic Programming Approach. *Theory and Practice of Logic Programming*, 1–49. doi:<https://doi.org/10.1017/S1471068426100441>.

Peer-reviewed Conference Proceedings

- C1. **Fossemò, Daniele** et al. *Paper on the foundations of Mechanistic Interpretability in Under review at a top-tier (A*) conference* (2026).
- C2. **Fossemò, Daniele**, Mignosi, F., Raggioli, L., Spezialetti, M., D'Asaro, F. A., et al. *Using inductive logic programming to globally approximate neural networks for preference learning: challenges and preliminary results* in *CEUR WORKSHOP PROCEEDINGS* (2022), 67–83. <https://ceur-ws.org/Vol-3319/paper7.pdf>.

Presentations

- T1. **Fossemò, Daniele**. *Using Inductive Logic Programming to Globally Approximate Neural Networks for Preference Learning: Challenges and Preliminary Results* 1st Workshop on Bias, Ethical AI, Explainability and the Role of Logic and Logic Programming (BEWARE 2022), co-located with AIxIA 2022 (University of Udine, Udine, Italy). Dec. 2022.

Tools & Software

Datasets

GRANITA (Gastronomic Recipes And Nutritional preferences from ITAly) is a dataset of Italian recipes paired with user preferences collected through surveys. Recipes are described by culinary attributes and by a hierarchical taxonomy of ingredients, while users express preferences on recipe features, together with pairwise preferences between recipes that also capture uncertainty, that is, pairs a user does not rank consistently. It is intended as a benchmark for preference learning algorithms and explainable AI methods alike, and underpins the experiments in *Explaining Neural Networks in Preference Learning: A Post Hoc Inductive Logic Programming Approach* (Theory and Practice of Logic Programming, 2026).

Last updated: October 1, 2026